

Math211 Midterm exam-Winter2016

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1. Consider this system of linear equations:

$$\begin{aligned} 3x + ky &= 1 \\ x - y &= 5 \end{aligned}$$

where k is a scalar, i.e., $k \in \mathbf{R}$. Choose the incorrect statement amongst the statements below.

- a) The system has no solutions when $k = -3$.
- b) The system has infinitely many solutions when $k = -3$.
- c) The system has a unique solution $x = \frac{5k+1}{k+3}$ and $y = -\frac{14}{k+3}$ when $k \neq -3$.
- d) The system has a unique solution when $k \neq -3$.
- e) The system cannot have infinitely many solutions.

2. Given the two matrices A and B given as

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 0 \end{pmatrix},$$

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}.$$

Which of these statements is incorrect?

- a) AB is not equal to BA since they have different sizes.
- b) BA is equal to I_2 .
- c) $A = B^T$.
- d) AB is equal to

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

- e) BA is not invertible.

3. Consider matrix

$$A = \begin{pmatrix} c & 1 & 2 \\ 1 & 0 & c \\ 2 & c & 0 \end{pmatrix},$$

where c is a real number. Determine when A is an invertible matrix.

- a) A is always invertible.
- b) A is invertible for every value of c except for $c = -2, 2$.
- c) A is invertible for every value of c except for $c = -2, 0, 2$.
- d) A is invertible for every value of c except for $c = 0, 2$.

e) A is invertible for every value of c except for $c = -2, 0$.

4. Let A , B and C be 4×4 matrices. If $\det A = 4$, $\det B = 2$ and $\det C = -1$, then $\det(\frac{1}{2}A^{-1}B^TC^3A^2)$ is equal to:

- a) 2
- b) -2
- c) $-\frac{1}{2}$
- d) $\frac{1}{2}$
- e) 1

5. Consider matrix

$$R = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}.$$

Which statement is correct?

- a) There exists a matrix B such that $RB = I_2$.
- b) R is invertible.
- c) The system of linear equations $RX = C$, has a unique solution.
- d) $R(\text{Adj}R) = (\text{Adj}R)R = (\det R)I_3$
- e) $R^{-1} = \frac{1}{\det R}\text{Adj}R$.